

There is a Fungus Among Us

2021 MCM Problem A: Fungi

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Abstract

Fungi are known to be parasitic organisms, taking energy from their hosts and decomposing them. One type of organism fungi feed on are trees; they do this by decomposing a tree's woody fibers and their ground litter (plant decay). As shown in multiple studies (including the one this project is based on), the decomposition rate of a fungi is dependent on a multitude of factors, such as air quality, temperature, the type of fungi, moisture, and more. In this project, we will explore two of these factors – temperature and moisture – and determining how these factors affect fungi decomposition of woody fibers and ground litter. Furthermore, this project also seeks to demonstrate how fungi compete with each other in different moisture and temperature levels.

Statement of Contribution

Christy worked on the majority of the presentation. Miranda worked on parts of the write-up, made the data easily accessible, and assisted Jason in the coding of the simulations. Jason worked on the programming and simulations. Ethan worked on the write-up, the model, some of the simulations, and parts of the presentation. Darren worked on the write-up and parts of the presentation.

Contents

I	Introduction	3
A	Problem Summary	3
II	Background	3
III	Model Assumptions	4
A	Competition	4
B	Measuring Decomposition	4
C	Recurrence Relation	4
IV	Model Development	5
A	Model Construction	5
A.1	Lotka-Volterra-Logistic Model	5
A.2	Decomposition Model	5
A.3	Matrix Model	7
B	Model Validation	8
V	Results	8
A	Validation	8
B	LVL Model	9
C	Heatmap Analysis	10
D	Correlation Factors	13
VI	Conclusion	14
A	Our Conclusions	14
B	Strengths	14
C	Weaknesses and Limiting Assumptions	14
D	Future Study	15
VII	Article	16

I. Introduction

Fungi play a critical role in the survival of terrestrial ecosystems by decomposing organic matter and subsequently recycling nutrients through the environment. Specifically, fungi are a key contributor of the carbon cycle, the process by which carbon is recycled by the decomposition of compounds, notably, woody fibers. In this study, we examine the growth rates of several fungi species as well as their moisture tolerance levels and temperature tolerance levels. We built a mathematical model of woody fiber decomposition using the fungi growth rate. To account for the different temperature and moisture adjustment, we used a simple linear model. We model the decomposition of woody fibers in a given patch of land and investigate the influence of the presence of fungi in the aforementioned patch of land.

A. Problem Summary

- Fungi growth is dependent on the temperature and moisture level of the environment
- Fungi also decompose woody fibers at different rates that are dependent on the temperature and moisture level
- Fungi compete for the same space when near each other

II. Background

As the primary decomposers of organic material in terrestrial ecosystems, fungi are important organisms that obtain their livelihood by taking energy from their hosts. One type of organism fungi often prey on are trees, in which they decompose wood in order to obtain their energy. In a study titled “A trait-based understanding of wood decomposition by fungi,” biologists Nicky Lustenhouwer, Daniel S. Maynard, Mark A. Bradford, Daniel L. Lindner, Brad Oberle, Amy E. Zanne, and Thomas W. Crowther studied the speed at which certain fungi decomposed ground litter (dead plant material) and woody fibers. In their study, they found multiple factors that determined how fast certain fungi decomposed these two materials.

Our goal for this project is to model the decomposition of ground litter and woody fibers made by fungi as well as simulate how one species of fungus would interact when they are in the presence of another species of fungus. We employ a logistic growth equation (1) to investigate this interaction. Here, r is the growth rate (mm per day), ρ is the current hyphal length, and ρ_{max} is the maximum size of the hyphae (i.e., the carrying capacity).

$$\frac{d\rho}{dt} = r * \rho \left(1 - \frac{\rho}{\rho_{max}} \right) \quad (1)$$

Once the fungi have successfully expanded along the surface in our model, we need to determine how much wood has decomposed. We create a separate model to track decomposition by multiple fungal species, and we will find the fungi species that had decomposed the most amount of wood in each simulation.

III. Model Assumptions

A. Competition

- Temperature and moisture stay constant throughout the simulation to simplify computations.
- The wood in all simulations is the same type of wood.
- The fungus's competitive ranking [1] represents the competition coefficient in the logistic growth model.
- The carrying capacity is equal to the (arbitrary) maximum length of the fungus' hyphae.
- Assume there is no predation on insects.
- The carrying capacity is proportional to the growth rate.
- We are assuming the intraspecies competitive coefficient is 1 (arbitrarily chosen). This is difficult to quantify and measure.
- Length is the variable being measured because hyphal growth rate was much more accessible than area.

B. Measuring Decomposition

- Assume there is no predation on insects.
- Assume that decomposition rate is related to hyphal extension rate.
- Assume the only variables affects the growth rate is the species, temperature of the environment and moisture of environment.
- Assume all fungi follows the summary in table S2 in [2] that

$$\log(\text{growth rate}) = \log(\text{decomposition rate})^{0.19} \quad (2)$$

C. Recurrence Relation

- Time is discrete.
- Temperature and moisture stay constant throughout the simulation to simplify computations.

- The wood being modeled on is uniformly dense and is represented by a square grid.
- The wood in all simulations is the same type of wood.
- The hyphae extend in 4 directions: up, down, left, and right, and cannot grow off the grid. This is based off of the concept of a random walk, a behavior that fungi tend to display [3].

IV. Model Development

A. Model Construction

We construct this model to accurately track the growth of fungi on a log. Each fungi species decomposes the wood at a different rate, which is determined by its response to temperature and moisture.

A.1. Lotka-Volterra-Logistic Model

Firstly, we explore the interactions between two fungal species with varying growth rates and competitive rankings. The primary factor of which fungi are competing for, in this instance, is space, which will be measured in millimeters and denoted in the model by the fungus' hyphal length. From (1), we may interpret the quantity $\frac{\rho}{\rho_{max}}$ as the space already used. Therefore, the quantity $1 - \frac{\rho}{\rho_{max}}$ is equivalent to the space that is not yet used.

In this model, we will induce interspecies competition through a competition coefficient p . We will investigate the interspecies as well as intraspecies interactions through a Lotka-Volterra model. However, instead of the classical model, we instead utilize a logistic growth equation (2). We call this model the Lotka-Volterra-Logistic model, or LVL model, for short.

$$\begin{aligned}\frac{d\rho_1}{dt} &= r \cdot \rho_1 \left(1 - \frac{\rho_1}{\rho_{1,max}} - p_2 \cdot \frac{\rho_2}{\rho_{2,max}} \right) \\ \frac{d\rho_2}{dt} &= r \cdot \rho_2 \left(1 - p_1 \cdot \frac{\rho_1}{\rho_{1,max}} - \frac{\rho_2}{\rho_{2,max}} \right)\end{aligned}\tag{3}$$

A.2. Decomposition Model

The second application of this model is showing the decomposition of the log by each fungi species. It will be noted which fungi species are decomposing faster while also accounting for their individual species growth on the log, as they should be related based on the data given. We will also be able to determine how the growth of a specific species can affect the log decomposition.

The data points on 1 represent the predicted decomposition rate and 2 represents the actual decomposition rate found on the appendix. Using standard

conditions 1MPa and 22°C . Limitations: This model holds temperature and moisture constant. By using a linear regression to find the coefficients and y-intercepts of temperature, moisture that is specific to each species of fungi, we obtain a new growth rate that can be fed into the new decomposition model at a different temperature and moisture. By manipulating the growth rate based on the equation, we could build different decomposition model at different temperature and moisture. However, the accuracy of the model will likely be lower.

$$growth\ rate = T \cdot k + t0 + m0 + M \cdot h \quad (4)$$

- T = temperature
- k = temperature constant, as temperature increases by 1 °C extension rate increases by k
- t0 = the hyphal extension rate at 0°C
- M= moisture (MPa)
- m= moisture constant, as moisture increases by 1 MPa, extension rate increases by k m0=the the hyphal extension rate at 0 MPa

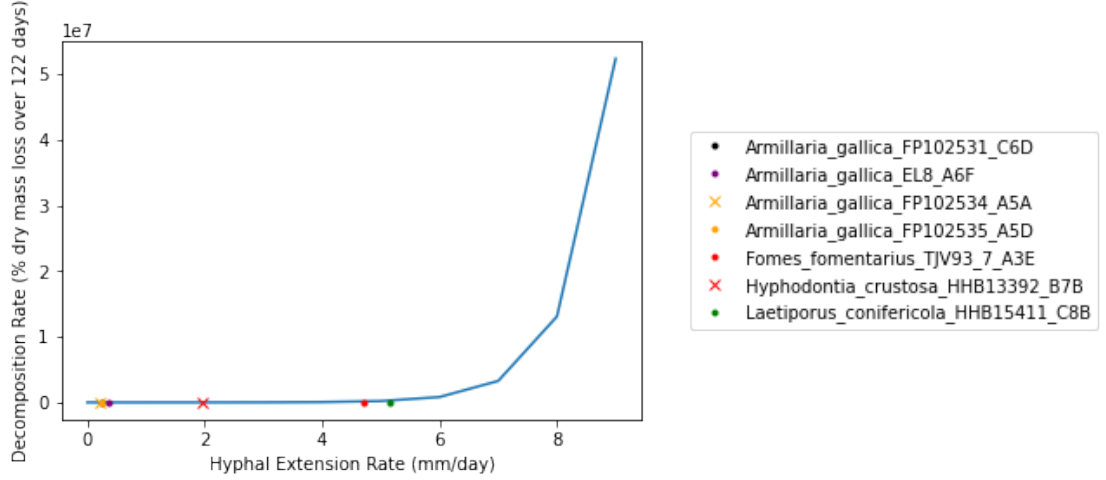


Figure 1: Predicted Decomposition Rate for Each Species

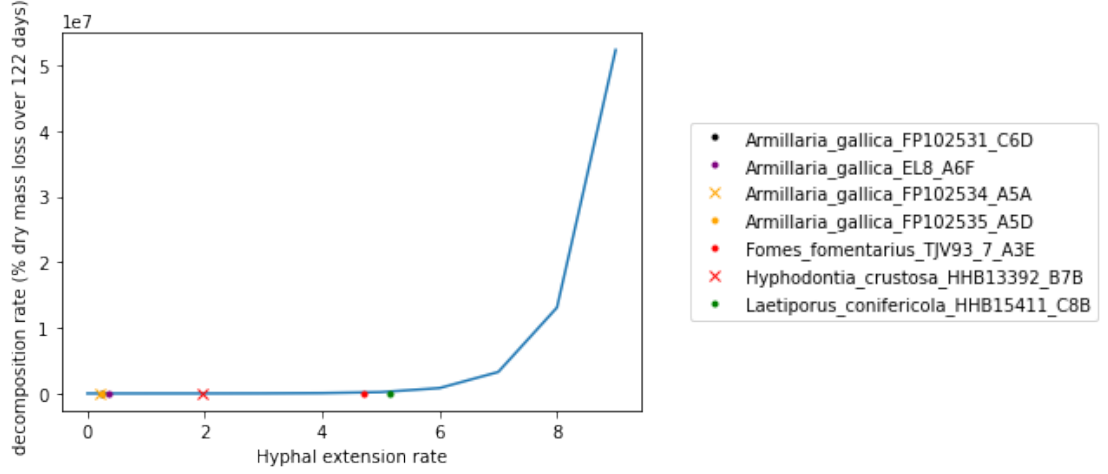


Figure 2: Actual Decomposition Rate for Each Species

A.3. Matrix Model

Thirdly, the model focuses on the growth of any given fungal species in seeing how they would grow in a 4 mm by 4 mm grid. From this we may draw inferences from any patterns of fungal growth that may be dependent on the temperature or moisture. We employ a transition matrix with given hyphal growth and expansion rates to track both the density and presence of the fungus in each 1 mm by 1 mm cell.

Multiplying the transition matrix by some initial starting point, x_0 , will give us the distribution, x_1 of the fungus throughout the grid in the next time step. This matrix multiplication can now be represented by a recurrence relation, where each time step gives us the distribution and density of the fungus in the grid. An example at time $t = 1$ is represented by Figure 3.

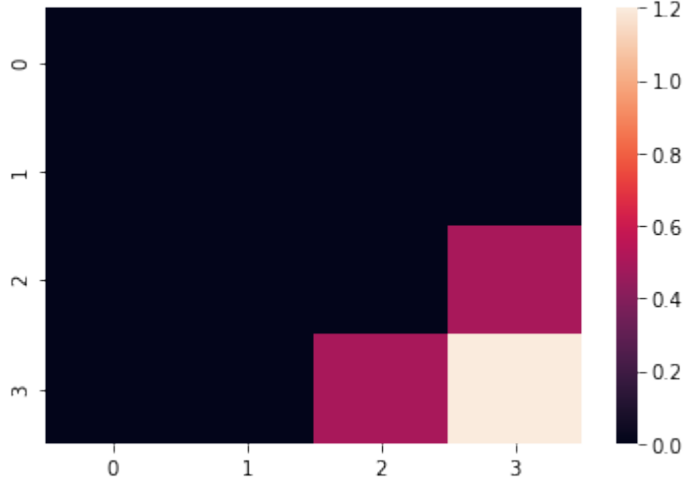


Figure 3: Example Heatmap at $t=1$

B. Model Validation

We will first validate our model qualitatively by following the growth path of each fungi species over time. We expect that the fungi will grow outwards in a circular pattern. We also expect the decomposition to follow a similar pattern to the growth of the fungi. Those that are expanding faster are expected to decompose the wood faster as well [2].

Quantitative expectations are hard to justify; however, we do expect that the fungi with higher growth rates will cover more of the log and decompose more of the log comparatively to the other fungi species. We expect these fungi to have higher competition rankings.

V. Results

A. Validation

As discussed in Model Validation, it is difficult to quantify our expectations of the model. As a result, many of the quantities in the models (particularly the Lotka-Volterra-Logistic model and the recurrence relation) had to be chosen arbitrarily. This has two inherent ramifications. One is that our models may be severely approximated or may grossly misrepresent reality. Another more positive one is that if more data is gathered, then these models could be adjusted or verified in the future.

B. LVL Model

We first investigate this model by exploring one fungus' interactions with another that has a competition coefficient of 0; we will also study their growth rates at 22 °C. We will begin with the initial condition of both lengths being 0.5 mm. The first fungus we are studying is *Armillaria gallica* HHB12551C6C (north), which has a growth rate of 0.48, a maximum length of 50 mm, and a competition ranking of 0. The fungus we are studying that interacts with this is *Armillaria gallica* FP102542A5B (south), which has a growth rate of 0.40, a maximum length of 45 mm, and a competition ranking of 0.3676. The results are given by Figure 4.

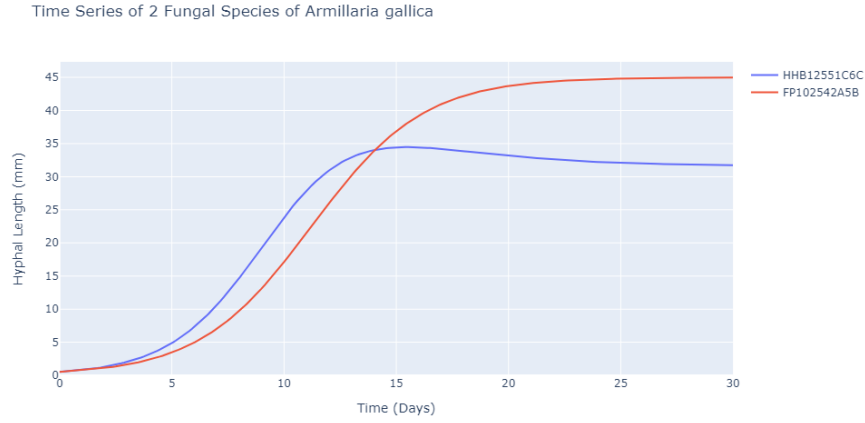


Figure 4: Graph with 1 Competition Ranking Being 0

From this model, we observe a that the species with a nonzero competitive ranking has a equilibrium hyphal length than the other.

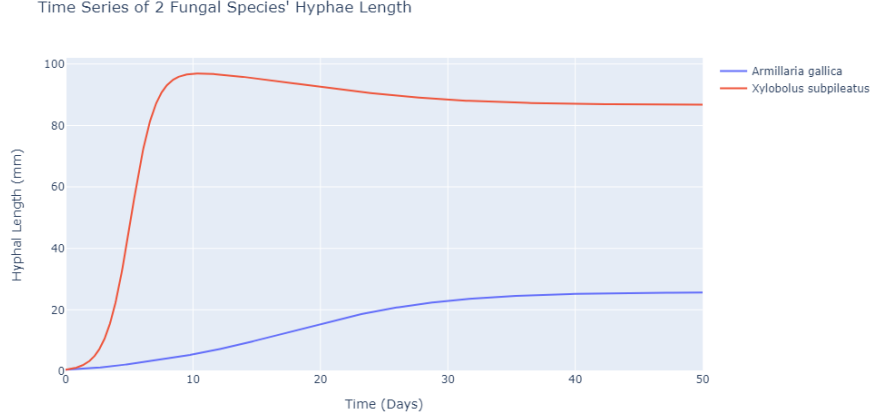


Figure 5: Graph with both Nonzero Competition Rankings

The next model we are investigating involved two species with nonzero competition rankings and growth rates at 22 °C. We will once again begin with the initial condition of both lengths being 0.5 mm. The first fungus we are studying is *Armillaria gallica* FP102531C6D(south), which has a growth rate of 0.34, a maximum length of 45 mm, and a competition ranking of 0.2325. The fungus we are studying that interacts with this is *Xylobolus subpileatus* FP102567A11A, which has a growth rate of 1.04, a maximum length of 100 mm, and a competition ranking of 0.2325. The results are given by Figure 5.

From this model, we observe that the species with a higher growth rate has a much larger equilibrium hyphal length than the other.

C. Heatmap Analysis

Figure 6 is a display of the heatmap at time steps 1, 5, and 10, with a growth rate of 0.34 mm/day at 22 °C (measurement drawn from *Armillaria gallica* FP102531C6D(south)). and an arbitrarily chosen expansion rate of 0.5 and 1, at the top and bottom of the figure, respectively.

Thus, depending on the expansion rate, we observe differences in how the density of the fungus is distributed across the log, represented by the grid.

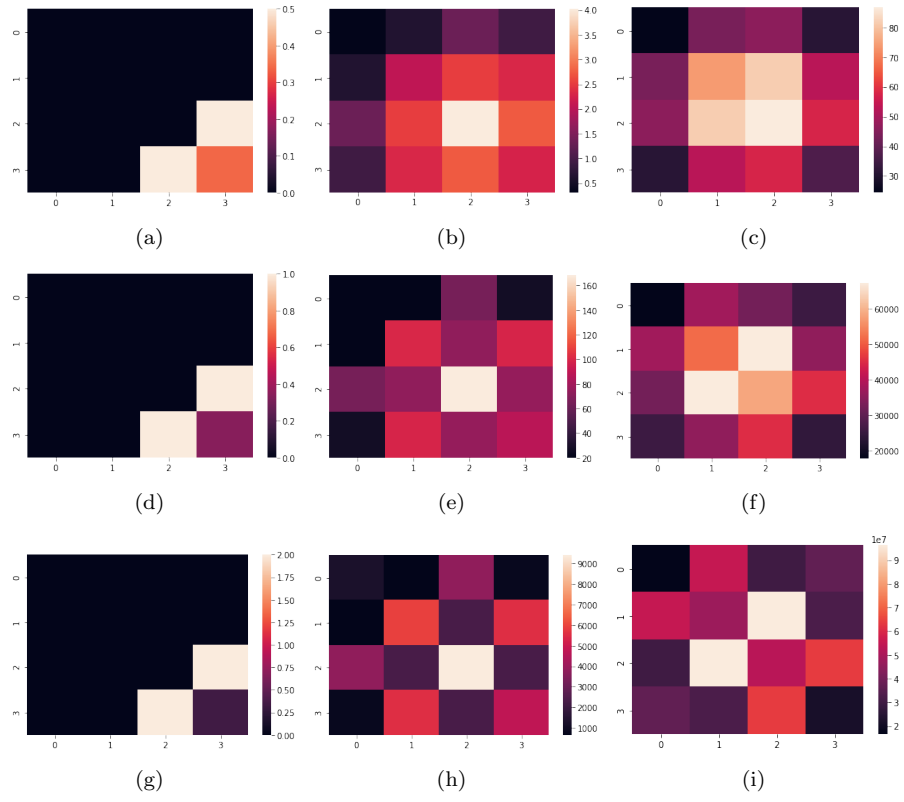


Figure 6: (a),(d),(h): $t = 1$ (b),(e),(i): $t = 5$ (c),(f),(j): $t = 10$

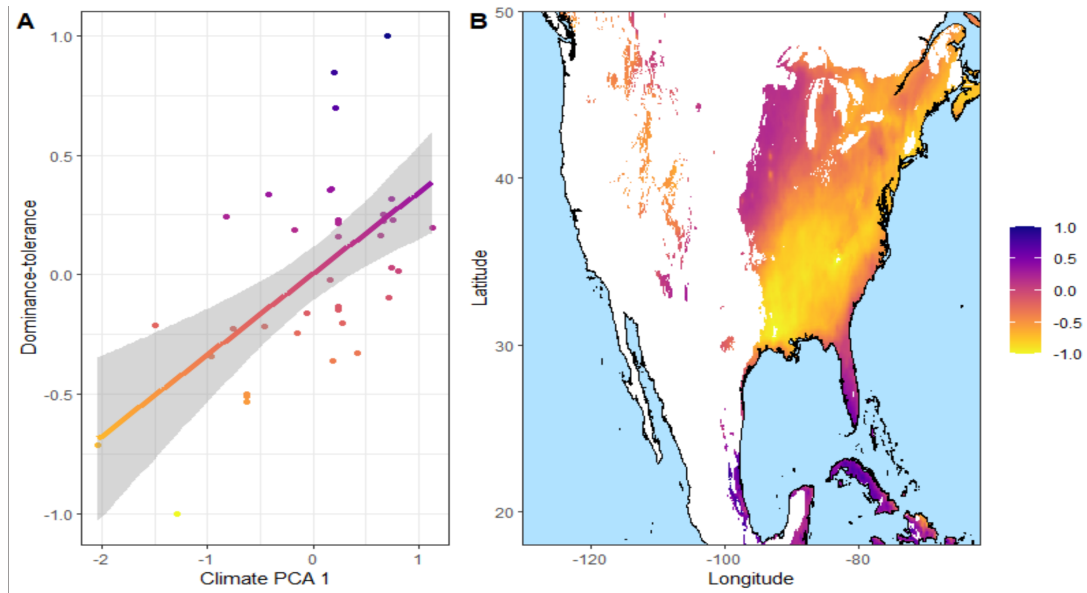


Figure 7: Dominance-tolerance Level of Fungi Mapped with Geographic Location

As shown in Figure 7, we can conclude that on the east coast of the US, the dominance tolerance of fungi decreases as the latitude increases (temperature decreases). Dominance tolerance of fungi are very high in regions where moisture and temperature are high, such as Florida and the coast of South America. The Midwest region also has a relatively high dominance tolerance rate. Thus, with this data, we infer that the dominance-tolerance is positively correlated to temperature and moisture.

D. Correlation Factors

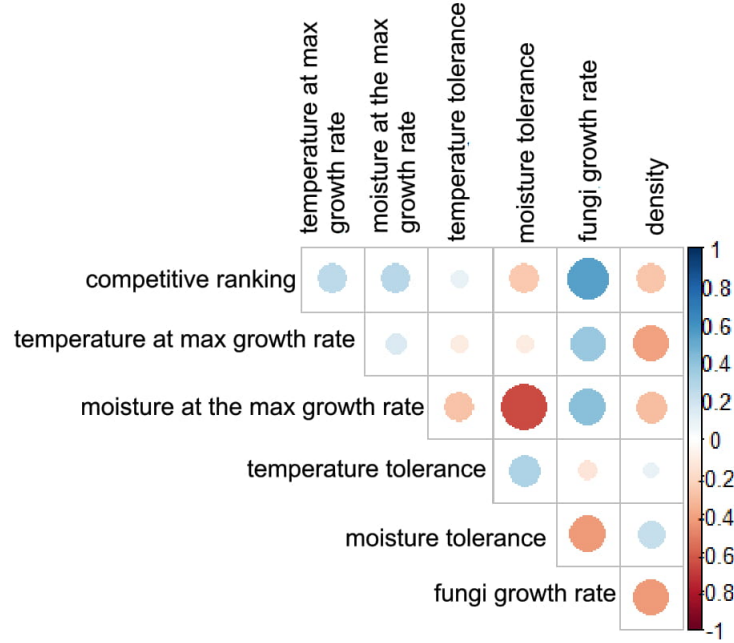


Figure 8: Correlation of Various Fungi Hyphal Extension Rates, Temperatures, and Moisture Levels

corrplot in R plots the correlation factor of competitive ranking, temperature at max hyphal extension rate, moisture at max hyphal extension rate, temperature tolerance, moisture tolerance and fungi growth rate. As shown in Figure 8, The competitive ranking of fungi is strongly positively correlated with fungi growth rate, moderately negatively correlated with density and moisture. The temperature at maximum growth rate is moderately positively correlated with fungi growth rate and moderately negatively correlated with density. In contrast, the moisture at maximum growth rate is strongly negatively correlated with moisture tolerance and moderately positively correlated with fungi growth rate. The temperature tolerance is moderately correlated to moisture tolerance, which implies the fungi species that has a higher temperature tolerance also tend to have a higher tolerance for moisture.

VI. Conclusion

A. Our Conclusions

- Our Lotka-Volterra-Logistic model allows us to observe the behavior of fungi hyphal growth when multiple fungi are situated in the same environment, given that the maximum length of the fungi are the limiting factors. Further study could involve linear stability analysis, among other methods of investigating this model.
- Through the exponential model of decomposition of fungi, we are able to use the hyphal extension rate of any given fungi species to predict the decomposition rate of the fungi at 22 °C and 1 MPa. Fungi species with a higher growth rate also breakdown a higher percentage of ground litter and woody fibers through fungal activity.

B. Strengths

- The model considers multiple different fungi and their interactions with one another, as opposed to a single fungi species expanding in one area without competition.
- The model uses a very realistic fungi growth that is drawn from experiments conducted with multiple runs.
- The Lotka-Volterra-Logistic model is easily applied to many different types of fungi that were not studied in this model.
- The matrix model lets us understand how fungi grow and expand in an environment with constant temperature and moisture.

C. Weaknesses and Limiting Assumptions

- The model ignores the use of multiple different types of wood which could decompose differently to different types of fungi under various conditions that what we had predicted
- The model only considers an arbitrarily small amount of wood, when in reality fungi can cover a much larger area.
- The model does not account for the death of fungi and how that could affect the decomposition rate of wood as well as the expansion rates into the area with dead fungi.
- The model is non-responsive to changes in temperature [4] or moisture and only models the growth and decomposition of fungi at set temperatures and moisture, when in reality temperature and moisture can change variably throughout the day or weeks [1].

- Many of the quantities chosen for the variables in the first and third models were purely arbitrary. Consequences of choosing arbitrary quantities is explained in Validation.
- The LVL model is severely limited in terms of dimensions; length is measured, instead of area, which would be a much stronger indicator of competition.

D. Future Study

There are many other factors beside temperature and moisture levels that influence the behavior of fungal growth. For example, certain fungal species utilize a range of sensory mechanisms as they develop in order to avoid obstacles, grow against gravity, and grow toward light and wind, among other things [5]. We could also investigate fungal growth in the presence of certain chemical compounds, and how those compounds influence the growth rate [6].

It should also be noted that we had only taken a few fungus species into account and having a larger range could possibly improve our model. Furthermore, when tracking the decomposition function and fungi growth, there was no account of death of fungi. If we are to account for this then we could have a better understanding of the different species and their effect on wood decomposition in the long term. This would also be helpful in understanding how the death of fungi would affect the ability for fungi to regrow in the same space.

VII. Article

Fungi are quite small and hard to determine their role in the global scale of the carbon cycle. Despite their ability to decompose woody fibers and partake in said cycle, it can be hard to quantify how important they have been. This also means that they lack proper data in showing how vital they have been. Through our models we believe to have achieved this goal in determining their worth in the carbon cycle.

Firstly our Lotka-Volterra model demonstrated competition and how the hyphal extension rate played a role in competition. What we learned was that those with higher hyphal extension rates were generally more competitive and exhibited greater growth amongst the two species. We also learned that varying the temperature and moisture heavily affected the equilibrium of the model. Those that were in hotter environments tended to exhibit faster hyphal extension rates. However once it had reached a certain temperature, this rate plummeted. Therefore, we see a weak correlation between the two factors. Finally, species with a high hyphal extension rate also tend to have a higher competition ranking among all the species.

Our Decomposition Model demonstrated that species with high hyphal extension rates were more likely to decompose woody fibers faster. Our model had accurately predicted the decomposition rate for various species of fungi and therefore we believe that it could be applicable to many more species that were not mentioned. What we can take from this is that, when modeling fungi growth, we can look to the hyphal extension rate of a species to determine the decomposition rate assuming a temperature of 22 °C. This also shows why this model might be limiting in terms of our conclusions since it only accounts for one temperature and moisture; however, we still believe that this demonstrates the relationship between the two factors.

Finally, the grid model better accurately depicts how exactly fungi are growing and competing with each other. This demonstrates the strong effects of temperature and moisture in the growth rate of different species of fungi. We were able to show just three different types of fungi but overall demonstrates that those that have higher hyphal extension rates (under the ideal conditions) are likely to grow faster and have denser spots on the grid which overall means more decomposition in each of those spaces. There is some regard in this conclusion as the competition amongst fungi is simply just the inability to inhabit the same space and in reality, won't happen exactly as we modeled.

In summary, our models consider various significant factors in modeling the behavior of fungal growth in various temperatures and moisture conditions. Through these considerations and our sensitivity analysis, we quantitatively demonstrated a relationship between growth rate and decomposition rate with respect to varying temperatures, moisture, and competition.

References

- [1] Bradford M.A. Covey K.R. et al. Maynard, D.S. *Consistent trade-offs in fungal trait expression across broad spatial scales*. Nat Microbiol 4, 846–853, 2019.
- [2] Maynard D.S. Bradford B.A. et al. Lustenhouwer, N. *A trait-based understanding of wood composition of fungi*. PNAS license, 2020.
- [3] S. Hopkins. *A Hybrid Mathematical Model of Fungal Mycelia: Tropisms, Polarised Growth and Application to Colony Competition*. University of Glamorgan, 2011.
- [4] Baath E. Rousk J. *Growth of saprotrophic fungi and bacteria in soil*. FEMS Microbiology Ecology, 2011.
- [5] David B. Dusenbery. *Life at Small Scale*. Scientific American Library, 1996.
- [6] A. Lamour. *Quantification of fungal growth: models, experiments, and observations*. 2002.